

An Automatic Following/Forming Framework for Inverter-Based Resources

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Problem

- 1 Converters are replacing synchronous machines, so **system inertia falls** and sensitivity to disturbances increases.
- 2 **GFL** regulates power using PLL-based synchronisation, which can lead to oscillations or instability in a weak or low-inertia grid.
- 3 **GFM** establishes its own voltage and frequency reference to support operation under these conditions.
- 4 Existing work **fixes** the switching condition and the number of forming units in advance, rather than adapting them to network conditions.

Introduction

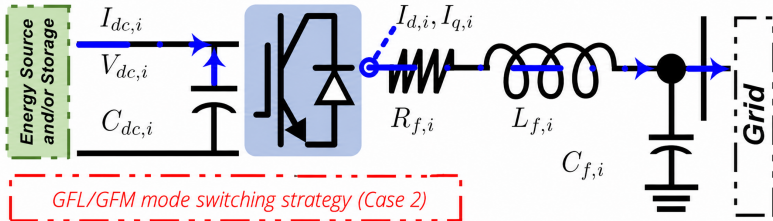


Figure 1: Electrical model of one inverter-based resource (IBR).

$$m_i(t) \in \{0, 1\}, \quad 0 = \text{GFL}, \quad 1 = \text{GFM} \quad (1)$$

$$\dot{\mathbf{x}}_{IBR,i} = \begin{cases} \mathbf{f}_{GFL}(\mathbf{x}_{GFL}, \mathbf{y}, \mathbf{u}, t), & m_i(t) = 0, \\ \mathbf{f}_{GFM}(\mathbf{x}_{GFM}, \mathbf{y}, \mathbf{u}, t), & m_i(t) = 1. \end{cases} \quad (2)$$

$\mathbf{x}$ converter and machine states
 $\mathbf{y}$ network bus voltages

$\mathbf{u}$ inputs: references, loads, fault status
 t time; events are applied at exact instants

Converter and Machine States

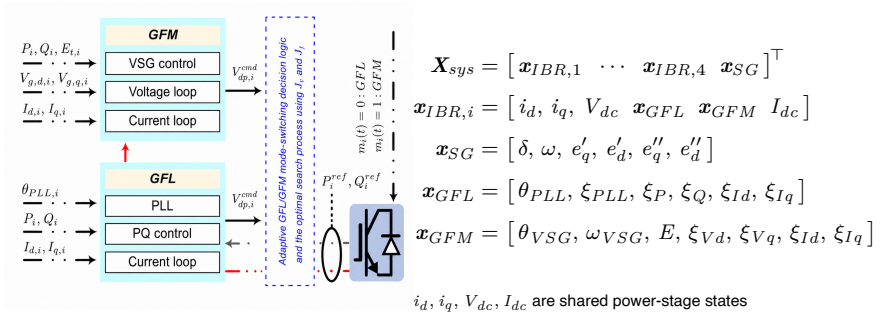


Figure 2: The two controller branches and the mode-switching decision.

i_d, i_q, V_{dc}, I_{dc} are shared power-stage states integrated in both modes. Only one of $\mathbf{x}_{GFL}, \mathbf{x}_{GFM}$ is integrated at a time.

Switching Decision

$$J_{V,i}(t) = \frac{||V_i(t)| - V_i^*|}{\Delta V_{base}} \quad (3)$$

$$J_f(t) = \frac{|f_{coi}(t) - f_0|}{\Delta f_{base}} \quad (4)$$

$$S_i(t) = \text{sat}_{[0,1]}(w_V J_{V,i} + w_f J_f) \quad (5)$$

$$0 \rightarrow 1 : S_i \geq \Gamma_{on} \text{ for } T_{d,on} \quad (6)$$

$$1 \rightarrow 0 : S_i \leq \Gamma_{off} \text{ for } T_{d,off} \quad (7)$$

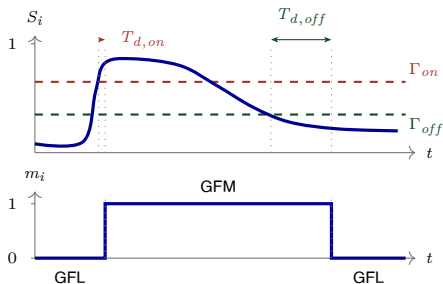


Figure 3: Mode switching on the severity index.

$J_{V,i}$ voltage deviation of bus i

J_f frequency deviation

V_i^* bus i 's healthy power-flow $|V|$

S_i severity of converter i , in $[0, 1]$

Γ switching threshold on S_i

T_d required duration beyond the threshold

$f_0 = 60 \text{ Hz}$ $\Delta V_{base} = 0.10 \text{ p.u.}$

$\Gamma_{on} = 0.65$ $\Gamma_{off} = 0.35$

$\Delta f_{base} = 0.50 \text{ Hz}$

$T_{d,on} = 0.10 \text{ s}$

$w_V = w_f = \frac{1}{2}$

$T_{d,off} = 1.00 \text{ s}$

Power Flow and Time-Domain Simulation

Power-flow operating point

$$\mathbf{0} = g(x, y, u_0, m) \quad (8)$$

Power must balance at every bus j :

$$(P_j^G + P_j^{inj}) - P_j^L - P_j(y) = 0 \quad (9)$$

$$(Q_j^G + Q_j^{inj}) - Q_j^L - Q_j(y) = 0 \quad (10)$$

Time-domain response

$$\dot{x} = f(x, y, u, m, t) \quad (11)$$

$$\mathbf{0} = g(x, y, u, m, t) \quad (12)$$

One step $t_n \rightarrow t_{n+1}$ of size h :

$$x_{n+1} = x_n + \frac{h}{2} [f_n + f_{n+1}] \quad (13)$$

$$\mathbf{0} = g_{n+1} \quad (14)$$

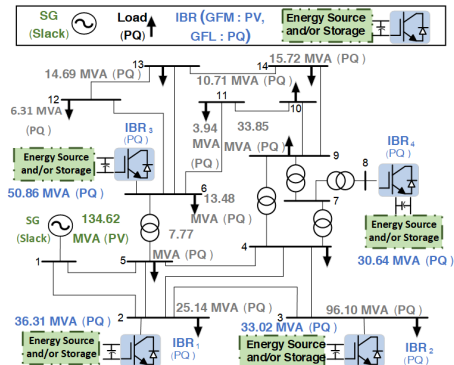
Symbols

P_j^G, Q_j^G machine injection at bus j
 P_j^{inj}, Q_j^{inj} **specified** converter injection
 P_j^L, Q_j^L load at bus j
 u_0 fixed inputs at the operating point
 $P_j(y), Q_j(y)$ network injection with fixed $Y_{bus}(u_0)$
 $|V_j|, \theta_j$ **calculated** bus voltage

Symbols

x converter and machine states
 y bus voltages
 $u(t)$ references, loads and fault status
 $m(t)$ converter modes that **change** over time
 h step size, $t_{n+1} - t_n$
 f_n, g_n (11)–(12) evaluated at $(x_n, y_n, u_n, m_n, t_n)$
 (13) applies to the integrated coordinates

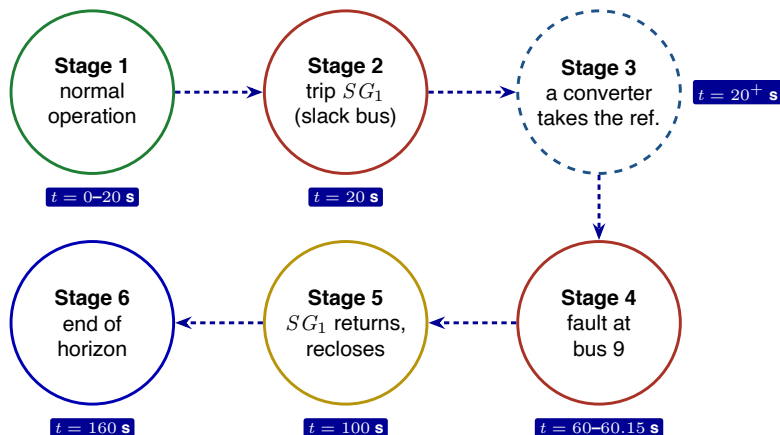
Case Study: IEEE 14-Bus System



Resource	Bus	MVA
SG (slack)	1	134.62
IBR ₁	2	36.31
IBR ₂	3	33.02
IBR ₃	6	50.86
IBR ₄	8	30.64

Figure 4: The IEEE 14-bus study network.

Scenario Timeline



Result: The Index and the Mode

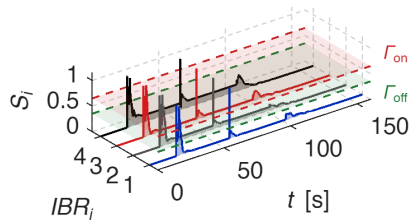


Figure 5: Switching index S_i per converter.

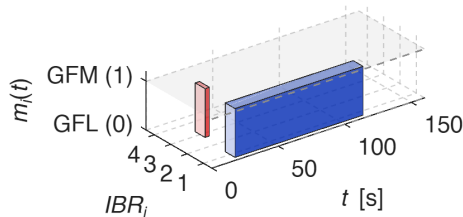


Figure 6: The resulting mode $m_i(t)$ per converter.

Result: Voltage and Frequency

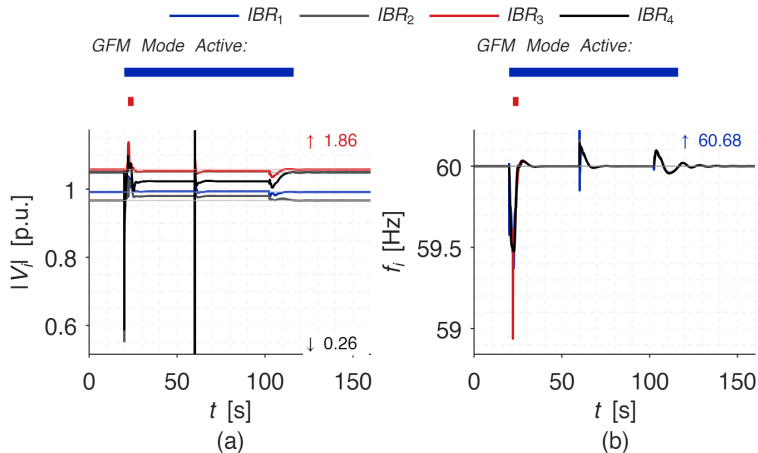


Figure 7: (a) $|V_i|$ at each converter bus and (b) each converter's own frequency f_i .

Result: Active and Reactive Power

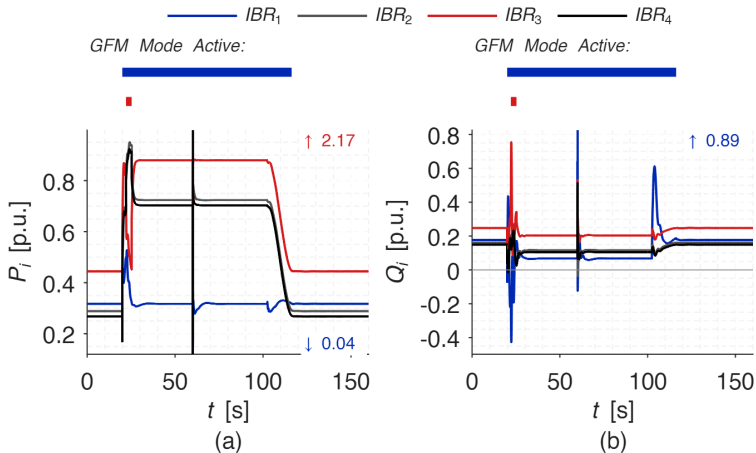


Figure 8: (a) P_i and (b) Q_i at each converter.

Comparison: Adaptive against Fixed GFL

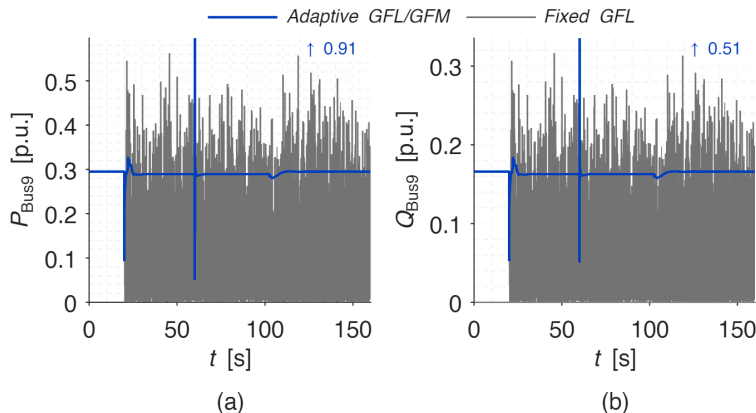


Figure 9: (a) P_{Bus9} and (b) Q_{Bus9} : adaptive (blue) against fixed GFL (grey).

Comparison: Adaptive against Fixed GFL

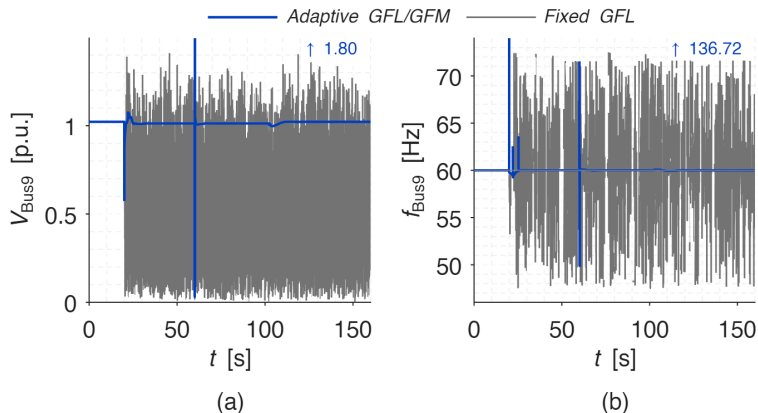


Figure 10: (a) V_{Bus9} and (b) f_{Bus9} : adaptive (blue) against fixed GFL (grey).

Proof: The Switched Dynamics

1. The same 17 states in both modes

$$\mathbf{x}_{IBR} = \begin{bmatrix} \mathbf{x}_p \\ \mathbf{x}_{GFL} \\ \mathbf{x}_{GFM} \end{bmatrix} \quad \begin{array}{l} 4 \text{ physical states: } i_d, i_q, V_{dc}, I_{dc} \\ 6 \text{ GFL states: PLL, PQ and current control} \\ 7 \text{ GFM states: VSG, voltage and current control} \end{array} \quad (15)$$

$$\begin{array}{cc} \text{GFL } (m = 0) & \text{GFM } (m = 1) \\ \dot{\mathbf{x}}_{IBR} = \begin{bmatrix} \mathbf{f}_p^{(0)}(\mathbf{x}, \mathbf{y}, \mathbf{u}, t) \\ \mathbf{f}_{GFL}^c(\mathbf{x}, \mathbf{y}, \mathbf{u}, t) \\ \mathbf{0}_7 \end{bmatrix} & \dot{\mathbf{x}}_{IBR} = \begin{bmatrix} \mathbf{f}_p^{(1)}(\mathbf{x}, \mathbf{y}, \mathbf{u}, t) \\ \mathbf{0}_6 \\ \mathbf{f}_{GFM}^c(\mathbf{x}, \mathbf{y}, \mathbf{u}, t) \end{bmatrix} \end{array} \quad (16)$$

$\mathbf{f}_p$: physical equations. $\mathbf{f}^c$: controller equations. $\mathbf{0}_6, \mathbf{0}_7$: inactive controller states stay constant — they are held, not solved, and are *re-initialised* rather than continued when the mode next becomes active.

Proof: The Switched Dynamics

2. At t_s : initial states for the new mode

The state is re-assigned, not integrated

$$x^+ = R(x^-, y^-, u^-, u^+), \quad g(x^+, y^+, u^+, m^+, t_s) = 0. \quad (17)$$

R sets the initial states for the chosen mode. $-$: just before t_s . $+$: just after t_s .

GFL to GFM: initialize the voltage-loop state

At unchanged V , choose $\theta^+ = \arg V$ and $E^+ = |V|$. Then $E^+ - v_d^+ = 0$.

$$\xi_{Vd}^+ = \frac{i_d^+}{k_{iV}} \implies i_{d,ref}^{raw,+} = k_p V(0) + k_{iV} \xi_{Vd}^+ = i_d^+, \quad k_{iV} \neq 0. \quad (18)$$

Likewise $\xi_{Vq}^+ = i_q^+ / k_{iV}$. These are matches of the *raw* reference, before current limiting.

Proof: The Switched Dynamics

3. Continuity and the post-switch transient

GFL/GFM switching with unchanged network and a locally regular DAE. On one current base, $i_{dq} = i_d + j i_q$ and $i_s = i_{dq} e^{j\theta}$. θ is the PLL or VSG angle.

The angle changes, but the physical current stays the same

$$i_{dq}^+ = i_{dq}^- e^{j(\theta^- - \theta^+)}, \quad V_{dc}^+ = V_{dc}^-, \quad I_{dc}^+ = I_{dc}^-. \quad (19)$$

$$i_s^+ = i_{dq}^+ e^{j\theta^+} = (i_{dq}^- e^{j(\theta^- - \theta^+)}) e^{j\theta^+} = i_s^-. \quad (20)$$

After t_s : integrate the equations of the new mode

$$\mathbf{x}(t_s + h) = \mathbf{x}^+ + \int_{t_s}^{t_s + h} \mathbf{f}_{m^+}(\mathbf{x}(\tau), \mathbf{y}(\tau), \mathbf{u}(\tau), \tau) d\tau. \quad (21)$$

$\mathbf{f}_{m^+}$ is the new mode's state derivative. If its magnitude stays below B for the applied $\mathbf{u}(\tau)$, then

$$\|\mathbf{x}(t_s + h) - \mathbf{x}^+\| \leq Bh \xrightarrow{h \rightarrow 0^+} 0. \quad (22)$$

The current is continuous, while its slope may change.
A transient can follow. Continuity alone does not prove stability.

Summary

Completed work

- ✓ Power flow and time-domain simulation
- ✓ GFL/GFM converter model
- ✓ Continuous current and DC states during mode transfer
- ✓ Switching based on voltage and frequency deviations
- ✓ ~~Selection of grid-forming converters~~
- ✓ ~~Comparison with fixed GFL control~~

Future work

- ☐ Stability conditions for mode switching
- ☐ Nonlinear stability analysis
- ☐ Systems with more converters
- ☐ EMT simulation, unbalanced operation and protection

The model combines GFL power regulation and GFM voltage and frequency support in one converter. Automatic mode selection allows the converter to respond to changing grid conditions.